

The equilibrium constant, K_c and magnitude of the K_c

Learning outcomes

By the end of this section you should be able to:

- (R2.3.2) Deduce the equilibrium constant equation from an equation for a homogeneous reaction.
- (R2.3.3) Relate the magnitude of the equilibrium constant for a reaction to the composition of the reaction mixture at equilibrium.
- (R2.3.3) Determine the relationships between the equilibrium constants for reactions that are the reverse of each other at the same temperature.

In the Big Picture (see [section R2.3.0 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43482/)), the concept of ‘extent of reaction’ was introduced. This can be thought of as how far to the right (the products side) a reaction proceeds. Different reactions proceed to different extents which means that some reactions proceed further to the right than others. But how do we know how far a reaction proceeds? Is there a way to quantify the concept of the extent of a reaction? The answer is yes and in this section you will learn about how the magnitude of the equilibrium constant, K_c , can be used as an indicator of the extent of a chemical reaction.

To begin this section, you will learn how to write the equilibrium constant expression for a reaction at equilibrium.

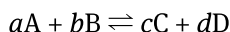
Study skills

The symbol for the equilibrium constant is usually written as K_c but can also be written as K . Be careful to distinguish this from the symbol used for the temperature in kelvin which has the symbol K but is not written in italics.

The equilibrium constant, K_c , of a reaction is a ratio that provides information about the composition of a reaction mixture at equilibrium. In other words, it gives information about the extent of the reaction. The concentrations of reactants and products in the equilibrium mixture are determined from experimental data.

To calculate the value of the equilibrium constant for a reaction, we need to first write an equilibrium constant expression for the reaction.

To do so, let's consider the general reaction between reactants A and B to form products C and D, where a , b , c and d are coefficients.



The equilibrium constant expression for this reaction is written as:

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

- Square brackets are used to represent the concentrations of the reactants and products in mol dm^{-3}
- The products are placed in the numerator and the reactants in the denominator.
- The concentrations of the reactants and products are raised to the power of their coefficients in the balanced equation.

State symbols are usually left out of the expression and the equilibrium constant is dimensionless (it has no units). It should also be noted that you are only required to write equilibrium constant expressions for homogeneous reactions, where all reactants and products are in the same state. This usually involves reactions where all species are present in the gaseous state (g), but can also include reactions where all species are present in the liquid state (l).

Let's look at the equilibrium constant expression for the Haber process in which nitrogen gas and hydrogen gas react to form ammonia.

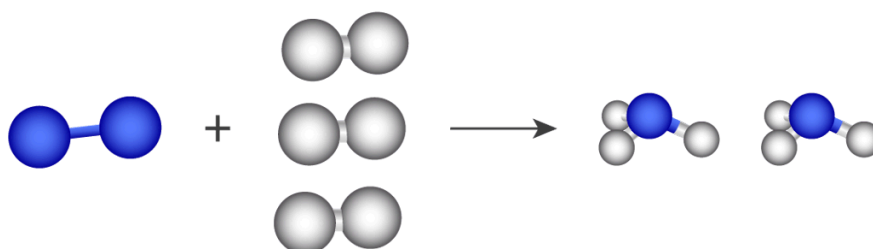
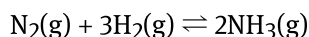


Figure 1. A molecular representation of the Haber process.

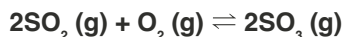
© More information for figure 1

The equilibrium constant expression for the reaction is:

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2] [\text{H}_2]^3}$$

Use **Interactive 1** to test your understanding of writing an equilibrium constant expression. Drag and drop the parts to the correct places in the expression.

Arrange the reactants and products into the correct places to construct the equilibrium constant equation for the reaction:



$$K_c = \underline{\hspace{2cm}}$$

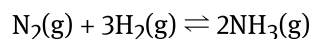


✓ Check

Interactive 1. Writing an equilibrium constant expression.

The magnitude of the equilibrium constant

If the equilibrium concentrations of reactants and products are known, the value of the equilibrium constant can be calculated. Take for example, the reaction between nitrogen and hydrogen to produce ammonia.



The equilibrium constant expression for the reaction is:

$$K_c = \frac{[\text{NH}_3]^2}{[\text{N}_2] [\text{H}_2]^3}$$

Table 1. The equilibrium concentrations of the reactants and products at a temperature of 500 °C

	Reactants		Products
	$\text{N}_2(\text{g})$	$\text{H}_2(\text{g})$	$\text{NH}_3(\text{g})$
Equilibrium concentration (mol dm^{-3})	0.80	2.40	0.40

Using this data, the equilibrium constant for the reaction can be calculated. To do this the equilibrium concentrations of the reactants and products are substituted into the equilibrium constant expression for the reaction:

$$\begin{aligned} K_c &= \frac{[0.40]^2}{[0.80][2.40]^3} \\ &= 0.01446\dots \\ &= 0.014 \text{ (2 s.f.)} \end{aligned}$$

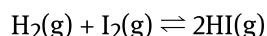
Now that we have calculated the equilibrium constant for the reaction at this temperature, what does this value tell us? It is relatively small, so do you think that there is a higher concentration of reactants or products at equilibrium? What if the value of the equilibrium constant is very large? The value of the equilibrium constant for a reaction tells us the extent of a reaction, or more specifically, to which side the equilibrium position lies. The equilibrium position can either lie to the left (reactants side) or to the right (products side). If the equilibrium position lies to the left, there is a greater concentration of reactants present in the equilibrium mixture. Conversely, if the equilibrium position lies to the right, there is a greater concentration of products present in the equilibrium mixture. This is summarised in **Table 2**.

Table 2. Equilibrium position.

Equilibrium position	Composition of equilibrium mixture
Lies to the right	Higher concentration of products
Lies to the left	Higher concentration of reactants

For the earlier example involving the Haber process, the value of the equilibrium constant was relatively small (0.0145) which means that at equilibrium, there is a higher concentration of reactants than products in the equilibrium mixture. So, for this reaction at this temperature, we say that the equilibrium position lies to the left.

In the next example, we have the reaction between hydrogen and iodine to produce hydrogen iodide.



Using the values in **Table 3**, calculate the value of the equilibrium constant.

Table 3. Equilibrium concentrations for the reaction of hydrogen and iodine.

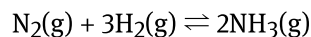
	Reactants		Products
	$H_2(g)$	$I_2(g)$	$HI(g)$
Equilibrium concentration (mol dm ⁻³)	0.0037	0.0037	0.028

$$\begin{aligned}
 K_c &= \frac{[HI]^2}{[H_2] [I_2]} \\
 &= \frac{[0.028]^2}{[0.0037] [0.0037]} \\
 &= 57.268 \dots \\
 &= 57 \text{ (2 s.f.)}
 \end{aligned}$$

What does this value of the equilibrium constant tell us about the composition of the reaction mixture at equilibrium? Compared to the previous example, this value of the equilibrium constant is considerably larger. So for this reaction at this temperature, there is a higher concentration of products than reactants. Therefore, we say that the equilibrium position lies to the right.

Worked example 1

At equilibrium, the reaction:



contains:

$$\begin{aligned}
 &0.30 \text{ mol dm}^{-3} H_2(g) \\
 &0.10 \text{ mol dm}^{-3} N_2(g) \\
 &0.80 \text{ mol dm}^{-3} NH_3(g)
 \end{aligned}$$

Determine the value of the equilibrium constant and state which side the equilibrium position lies.

$$\begin{aligned}
 K_c &= \frac{[NH_3]^2}{[N_2] [H_2]^3} \\
 &= \frac{[0.80]^2}{[0.10] [0.30]^3} \\
 &= 237.037 \dots \\
 &= 240 \text{ (2 s.f.)}
 \end{aligned}$$

The equilibrium constant is much greater than one, so the equilibrium position lies to the right and the mixture contains mostly products.

Figure 2 shows how the value of the equilibrium constant can be used to determine the composition of the equilibrium mixture for a reaction at equilibrium.

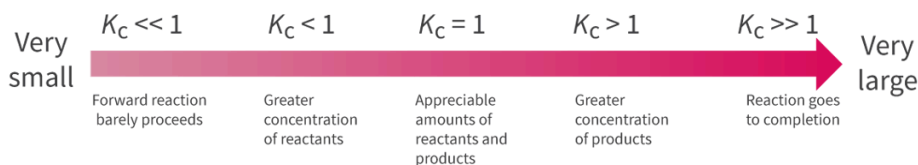


Figure 2. The value of the equilibrium constant is used to determine the composition of the equilibrium mixture.

📄 More information for figure 2

Table 4. A summary of equilibrium constant values and the composition of the reaction mixture

Value of K_c	Composition of reaction mixture
$K_c \ll 1$	Almost exclusively reactants
$K_c < 1$	Higher concentration of reactants
$K_c = 1$	Appreciable amounts of reactants and products
$K_c > 1$	Higher concentration of products
$K_c \gg 1$	Almost exclusively products

Making connections

How does the value of K for the dissociation of an acid convey information about its strength? [subtopic R3.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/)

Now try the drag and drop activity in **Interactive 2** to test your understanding of equilibrium constant values and the composition of the reaction mixture.

K_c value	Composition of the reaction mixture	Equilibrium position
4.61×10^{-3}		
640		
1.9×10^{-23}		
3.2×10^{81}		

Lies to the left

Forward reaction barely proceeds

Lies to the right

Almost exclusively products

Almost exclusively reactants

Higher concentration of products

Reaction goes to completion

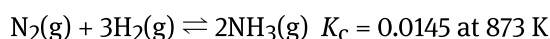
Higher concentration of reactants

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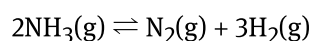
Interactive 2. Equilibrium constant values and the composition of the reaction mixture.

Determine the equilibrium constant for the reverse reaction at the same temperature

Earlier in the section, we looked at the equation for the Haber process and its equilibrium constant at 873 K.



But what about the equilibrium constant for reverse reaction at the same temperature?



The equilibrium constant equation for the reverse reaction is:

$$K_c = \frac{[\text{N}_2][\text{H}_2]^3}{[\text{NH}_3]^2}$$

Next, input the equilibrium concentrations of reactants and products into the

$$K_c = \frac{[0.80][2.40]^3}{[0.40]^2}$$

equation and solve for K_c .

$$= 69.12$$

$$= 69 \text{ (2 s.f.)}$$

Note that this is the reciprocal of the equilibrium constant for the forward reaction.

$$K_{c \text{ reverse}} = \frac{1}{K_{c \text{ forward}}}$$

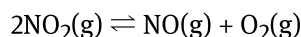
$$= \frac{1}{0.01446\dots}$$

$$= 69.12$$

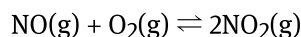
$$= 69 \text{ (2 s.f.)}$$

Worked example 2

The equilibrium constant for the reaction shown is 1.22×10^6 at a specific temperature.



Determine the equilibrium constant for the reaction below at the same temperature.



The reaction is the reverse of the original, so we can take the reciprocal of the equilibrium constant for the forward reaction.

$$K_{c \text{ reverse}} = \frac{1}{K_{c \text{ forward}}}$$

$$= \frac{1}{1.22 \times 10^6}$$

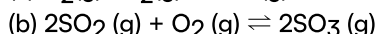
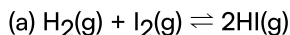
$$= 8.196\dots \times 10^{-7}$$

$$= 8.20 \times 10^{-7} \text{ (3 s.f.)}$$

Activity

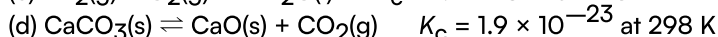
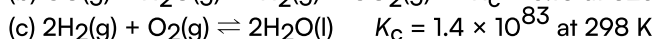
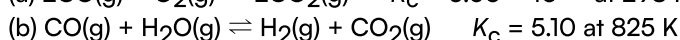
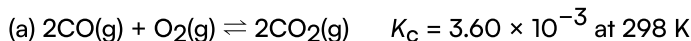
- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

1. Deduce the equilibrium constant equation for these reactions.



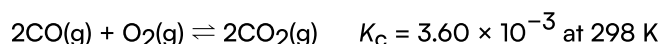
2. Outline what is meant by the term homogeneous reaction.

3. For each of the reactions below, determine the equilibrium position and the composition of the reaction mixture.

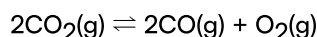


4. Explain the relationship between the magnitude of the equilibrium constant, the equilibrium position and the composition of the equilibrium mixture.

5. Carbon monoxide reacts with oxygen to produce carbon dioxide as shown.



Deduce the equilibrium constant for the following reaction at the same temperature.



1. (a) $K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]}$

(b) $K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]}$

2. The term homogeneous reaction refers to a reaction in which all the reactants and products are in the same state. It is also known as homogeneous equilibria.

3. (a) The equilibrium constant for this reaction is less than one ($K_c < 1$) so the equilibrium position lies to the left and the equilibrium mixture will contain a higher concentration of reactants.
 (b) The equilibrium constant for this reaction is greater than one ($K_c > 1$) so the equilibrium position lies to the right and the equilibrium mixture will contain a higher concentration of products.
 (c) The equilibrium constant for this reaction is much greater than one ($K_c \gg 1$) so the reaction goes to completion and the equilibrium mixture will consist of almost exclusively products.
 (d) The equilibrium constant for this reaction is much less than one ($K_c \ll 1$) so the forward reaction barely proceeds and the equilibrium mixture will consist of almost exclusively reactants.

4. The greater the value of the equilibrium constant, the further to the right the equilibrium position lies and the higher the concentration of products in the equilibrium mixture.

The smaller the value of the equilibrium constant, the further to the left the equilibrium position lies and the higher the concentration of reactants in the equilibrium mixture.

5. The equilibrium constant for the reverse reaction is the reciprocal of the equilibrium constant of the forward reaction.

$$\begin{aligned}K_c &= \frac{1}{3.60 \times 10^{-3}} \\&= 277.777\dots \\&= 278 \text{ (3 s.f.)}\end{aligned}$$